

**BACHELOR OF SCIENCE EXAMINATION, 2024**

**(1<sup>st</sup> Year, 2<sup>nd</sup> Semester )**

**MATHEMATICS**

**PAPER : SEC-02**

**[ Numerical Methods (Theory) ]**

*Time : 1 Hour 15 Minutes*

*Full Marks : 24*

Answer *any three* questions.

1. (a) The velocity  $v$  m/s of a particle at a time  $t$  s is given in the following table : 3+5

|     |   |   |    |    |    |    |     |
|-----|---|---|----|----|----|----|-----|
| $t$ | 0 | 2 | 4  | 6  | 8  | 10 | 12  |
| $v$ | 4 | 6 | 16 | 34 | 60 | 94 | 133 |

Find the distance travelled by the particle in 12 s.

**[CO3]**

- (b) Derive the iteration formula

$$x_{n+1} = \frac{1}{2} \left( x_n + \frac{a}{x_n} \right), \quad n = 0, 1, 2, 3, \dots$$

to find the square root of  $a$  ( $a > 0$ ), where  $x_0 (> 0)$  is any initial approximation and  $x_n$  is the  $n$ th approximation. Hence find the value of  $\sqrt{2}$  correct up to three significant figures. **[CO2]**



2. (a) Define round-off error and truncation error with proper examples. [CO1] 3+5
- (b) Find a formula

$$\int_0^1 \frac{f(x)}{\sqrt{x(1-x)}} dx = Af(0) + Bf\left(\frac{1}{2}\right) + Cf(1)$$

which is exact for polynomials of highest possible degree.  
Hence evaluate

$$\int_0^1 \frac{dx}{\sqrt{x(1-x^2)}} \quad [\text{CO2}]$$

3. (a) What are Lagrange's polynomials? How are they used for polynomial interpolation? [CO2] 3+5
- (b) Compute  $y(0.4)$ , from the equation

$$\frac{dy}{dx} = x - y, \quad y(0) = 1$$

taking  $h = 0.1$ , by using 4th order Runge-Kutta method, correct to five decimal places. [CO4]

4. (a) Give the geometrical interpretation of the Euler's method. [CO3] 3+5
- (b) Determine  $\alpha, \beta, \gamma$  and  $\delta$  such that the relation

$$y'\left(\frac{a+b}{2}\right) = \alpha y(a) + \beta y(b) + \gamma y''(a) + \delta y''(b)$$

is exact for polynomials of as high degree as possible. [CO4]



( 3 )

5. (a) Which one between the Secant method and the Regular-Falsi method has a higher rate of convergence? Explain why. [CO2] 4+4
- (b) Determine the step size  $h$  in a table of equally spaced values of the function  $f(x) = \sqrt{x}$  in  $[1,2]$ , so that the error in interpolation with a second degree polynomial in this table is less than  $5 \times 10^{-8}$ . [CO2]

